

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

THERMAL PROPERTIES OF MATTER

LECTURE-04 **PHYSICS**

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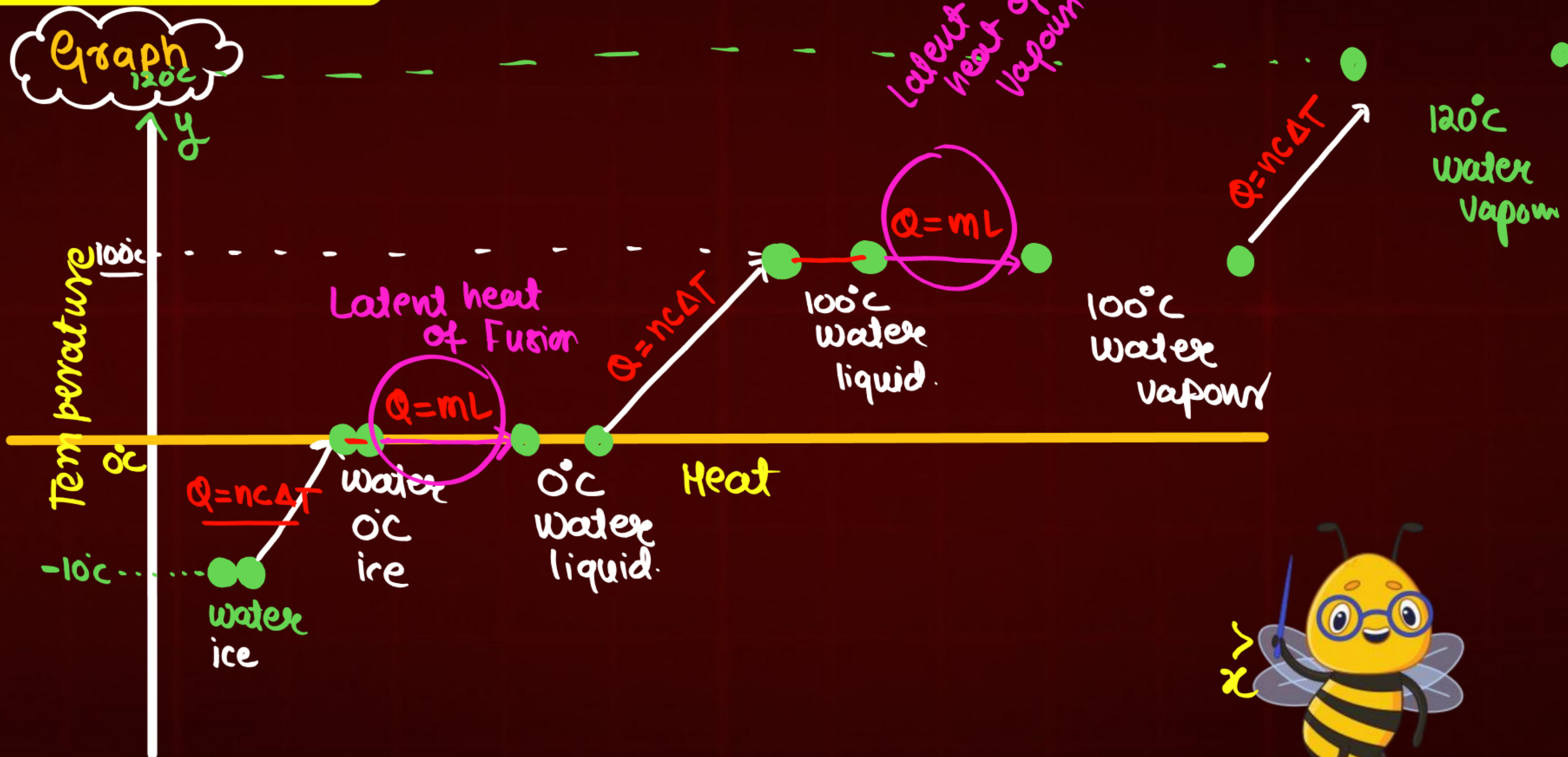
Topics To Be Covered

- 1** Revision
- 2** Calorimetry
- 3** Thermal Conductivity
- 4** Mode of Transfer of Heat
- 5** Newton Law of Cooling

Calorimetry

lecture no. 4

①



Thermal Conductivity

The Rate of flow of heat is called as Thermal conductivity

$$\text{Thermal conductivity} = \frac{\text{flow of heat}}{\text{time}} \quad (2)$$

$$\text{Thermal conductivity} = \frac{dQ}{dt}$$



$$\text{Thermal conductivity} = \frac{Q}{t}$$



Mode of Transfer of Heat

- How the heat is transferred from one place to another place.

2022

2023

2024

2026

3M

5M

③

mode of transfer of heat

1

Conduction

2

Convection

③

Radiation

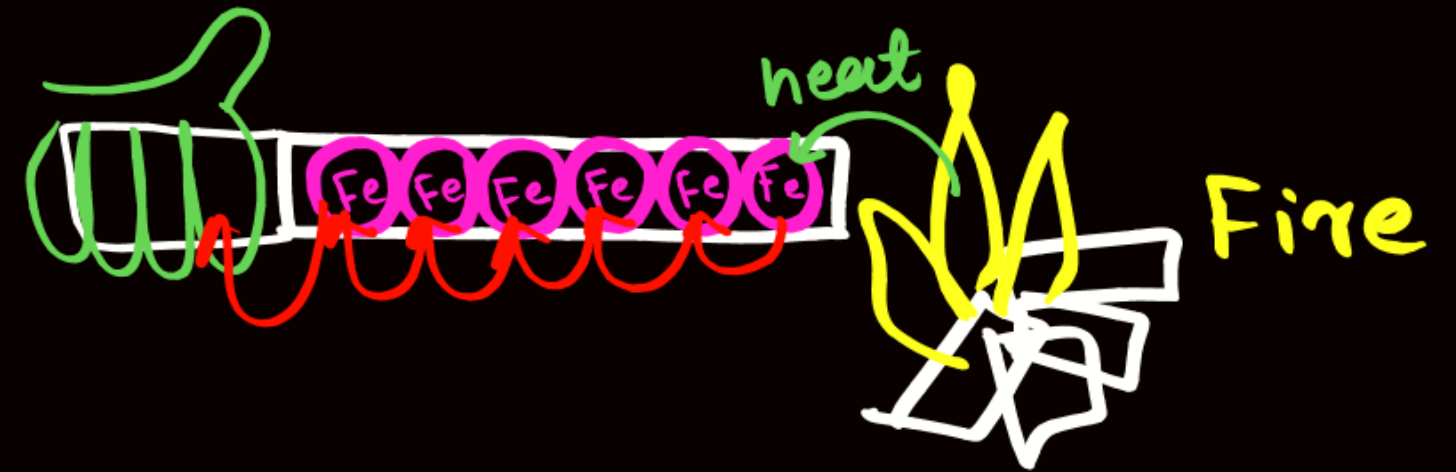


①

Conduction

- It is the mode of heat transfer
- In which material medium transfers the heat
- Without the movement of medium particle.

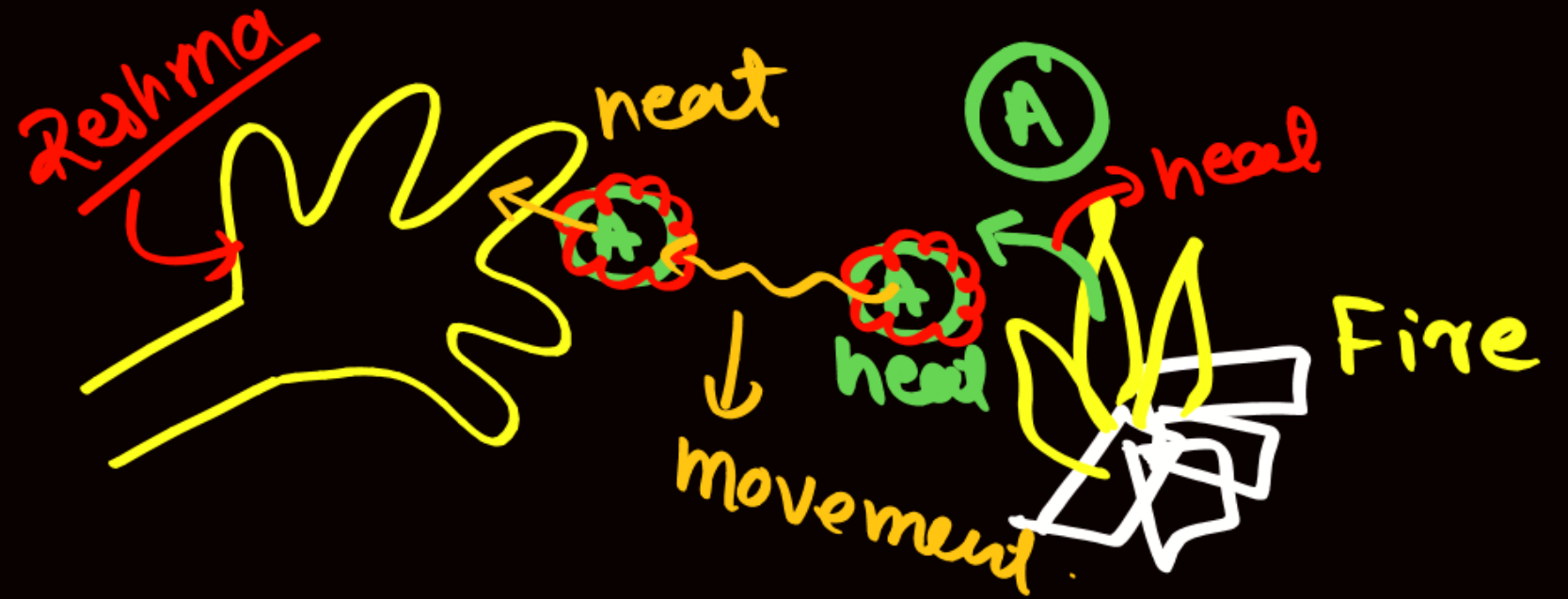
④



② Convection

④

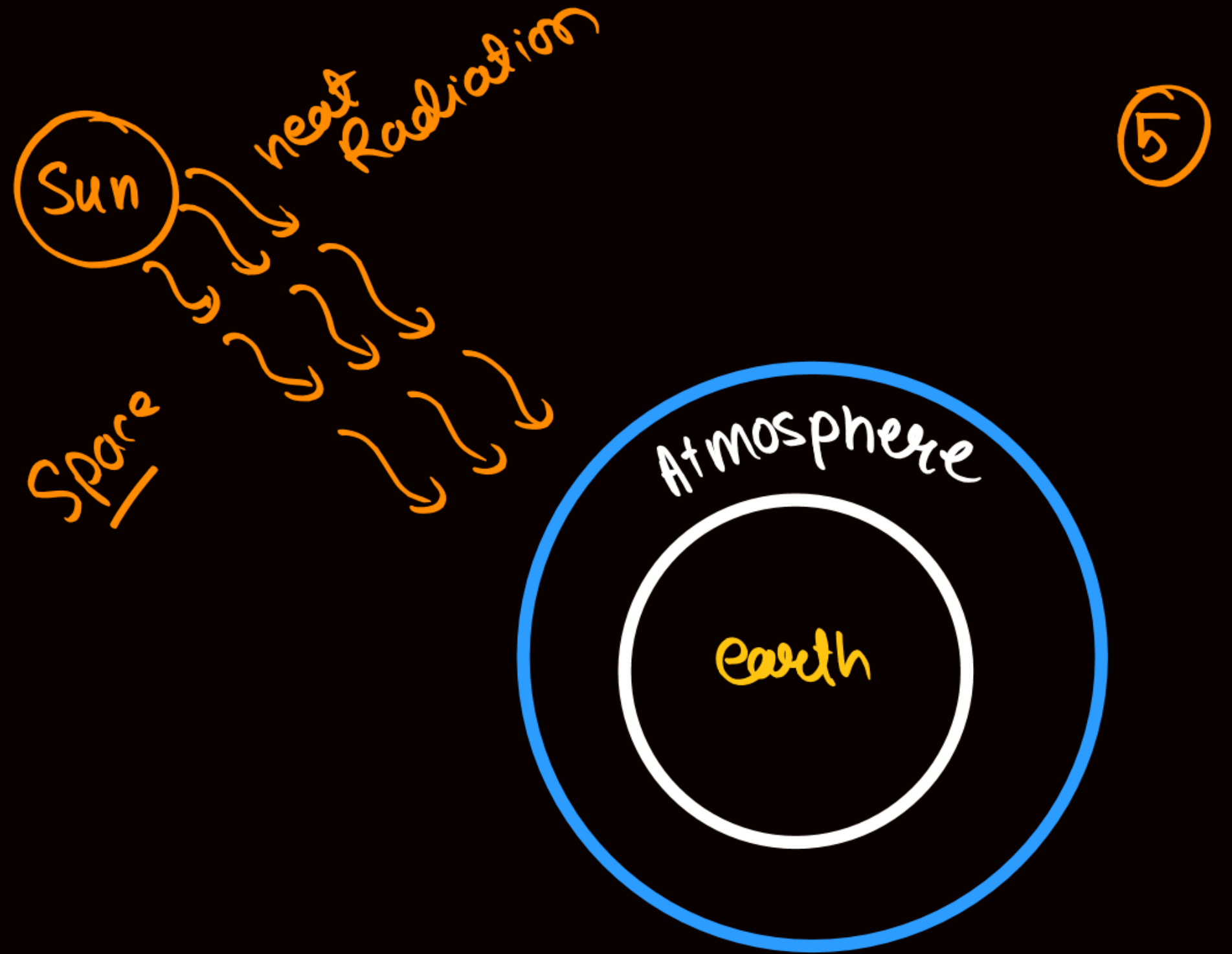
- ✓ It is the mode of heat transfer
- ✓ In which material medium transfers the heat
- ✓ With the movement of medium particle.



③ Radiation

- It is one of the mode of heat transfer

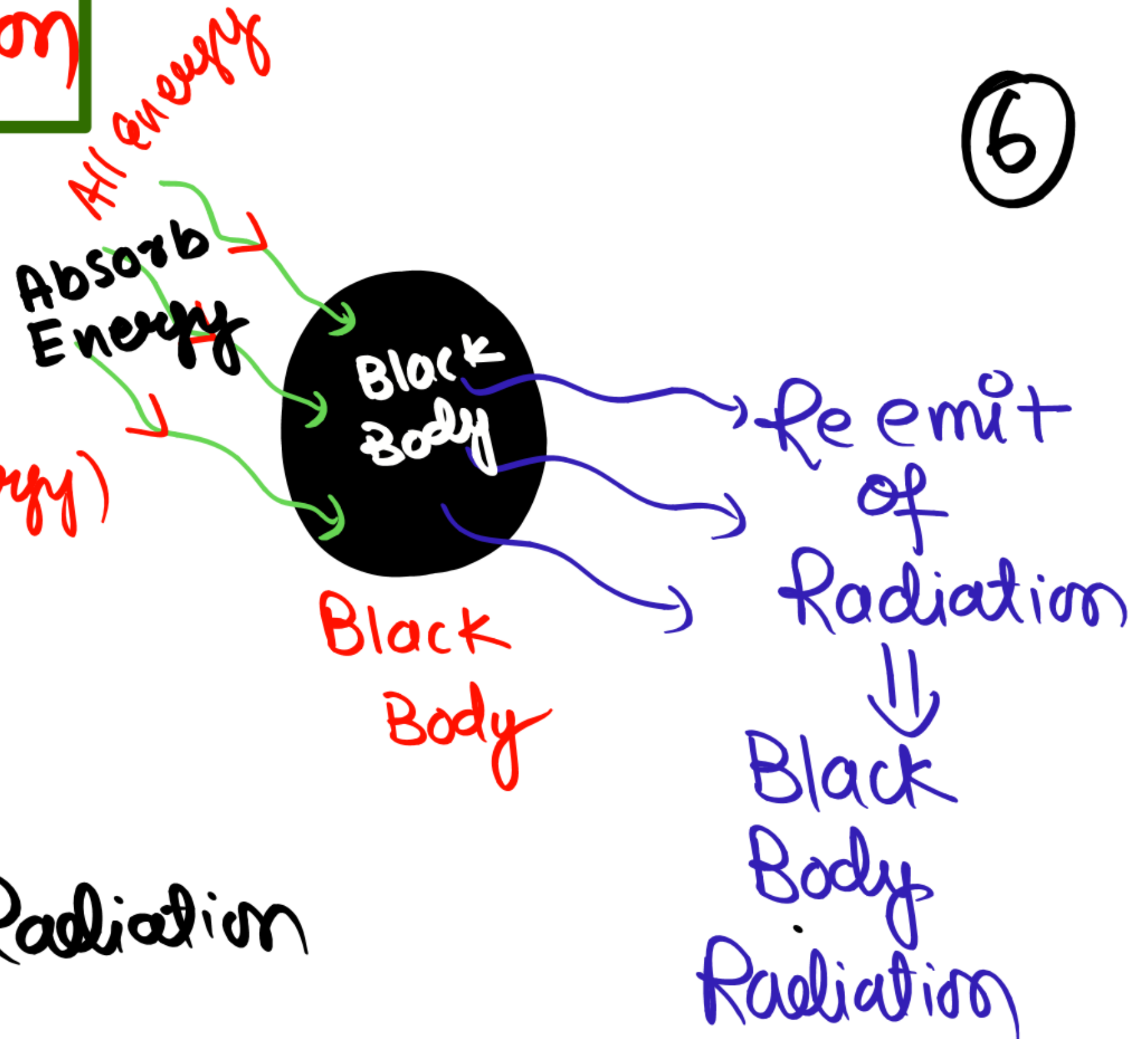
• no medium is Required.



Black Body Radiation

6

- # Black Body Absorb all type of Radiation (All Energy)
- # Once it is full
- # It Reemit the Radiation This is known as Black Body Radiation





Kirchoff law of Black Body Radiation

7

It States that

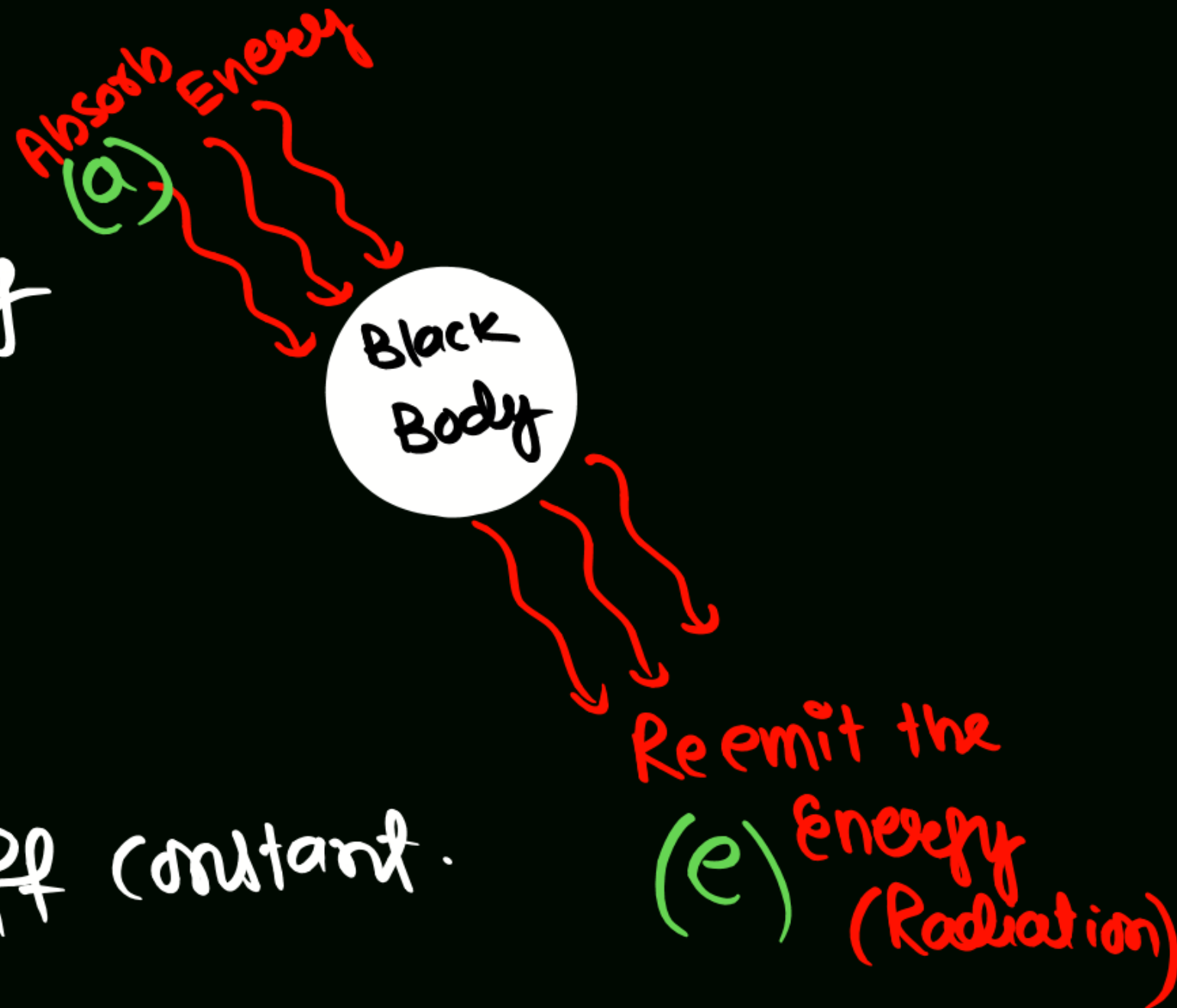
Absorbed Energy $\propto$ Reemit Energy

Absorption $\propto$ Emission

$$a \propto e$$

$$a = c e$$

$\rightarrow$ Kirchoff constant.



Newton law of cooling

3M/2M

- 4:00pm
- Newton chacha → pee Rake the → noon chai
- It states that the Rate of heat loss (Rate of cooling) is directly proportional to change in temperature

Rate of heat loss $\propto \Delta T$

$$\frac{dQ}{dt} \propto \Delta T$$

$$\frac{dQ}{dt} = (K) \Delta T$$

$K \rightarrow$ Newton cooling constant

Case I
heat loss (Fast) ⑧



$$T_s = 10^\circ C$$

$$\begin{aligned} \Delta T &= T_c - T_s \\ &= 80 - 10 \\ &= 70^\circ C \end{aligned}$$

Case II



$$T_s = 10^\circ C$$

$$\begin{aligned} \Delta T &= T_c - T_s \\ &= 15 - 10 = 5^\circ C \end{aligned}$$

Stefan law

2023 - 2 mark

9

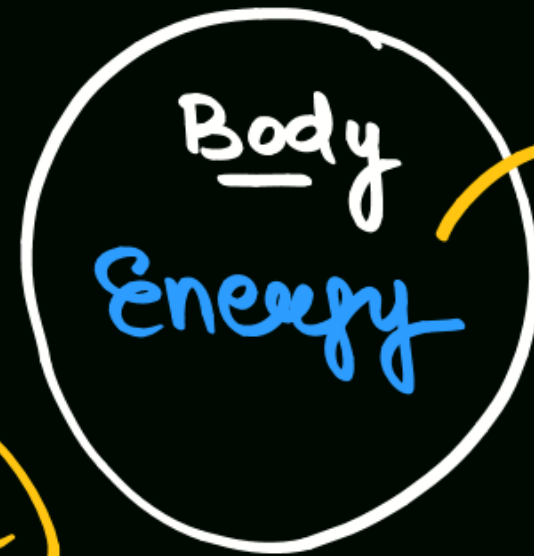
⇒ The Energy of the Body
is directly Proportional
to the fourth power of
temperature

Energy $\propto$ Temp⁴ (BRC)

$$E \propto T^4$$

$$E = \underline{k} T^4$$

k ⇒ Stefan Constant



→ Temperature

Wein displacement law

2026/2024/2022/2021/2018

~~Proof~~ Stefan law

Energy $\propto$ Temp

$$E \propto T$$

$\Rightarrow$ Planck Quantum theory

$$E = h\nu$$

$$c = \lambda\nu$$

$$\frac{c}{\lambda} = \nu$$

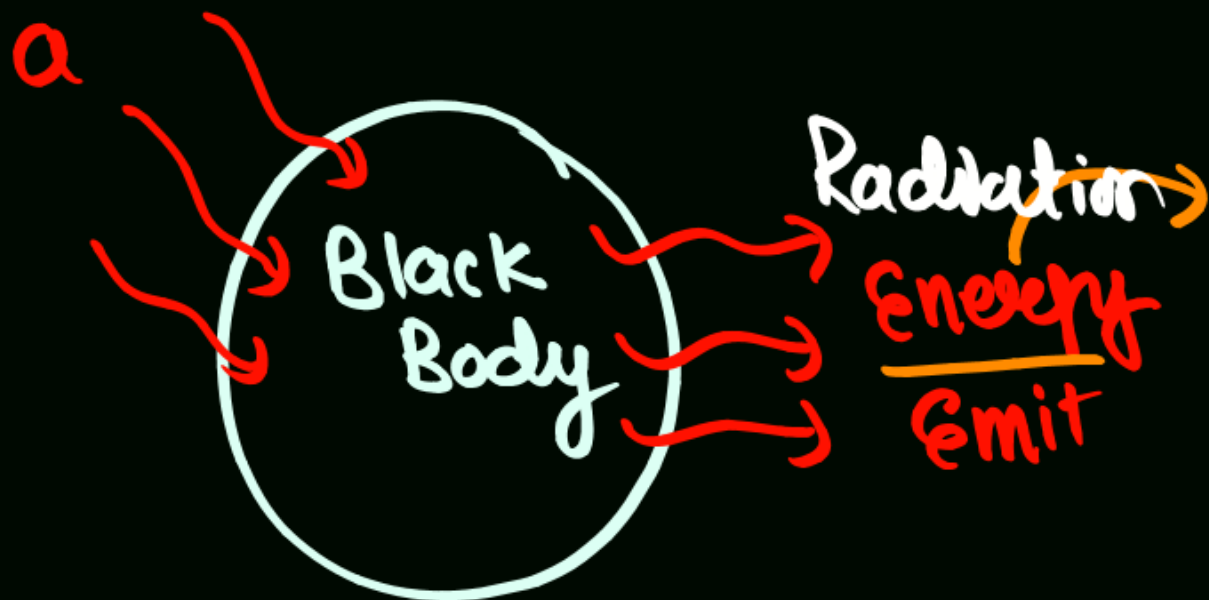
$$E = \frac{hc}{\lambda}$$

$$E \propto T$$

$$\frac{hc}{\lambda} \propto T$$

$$\lambda \propto \frac{1}{T}$$

$$\lambda \propto \frac{1}{T}$$



SUMMARY



q-c

Tc

MTH

C

C

R

BBR

KL

NL of cooling

SL

WDL

HOMework

Q what is water equivalent (Neet-2021)

Ans

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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Thank You

Revise
Lecture
Calculus
numerical
Series

